

Three policy targets, three policy levers: a stylised agent-based model in which household-sector inequality, productive-flow share, and population survival are addressed by distinct fiscal instruments

[Author]

(supervision and acknowledgements TBD)

Draft of June 1, 2026 for JASSS

Abstract

We present an agent-based model of an economy with seven occupationally heterogeneous agent classes and use it to argue that the fiscal-policy dimensions a wealth-tax debate typically lumps together — the rate schedule, the spending package, and the inheritance regime — act on distinct outcome metrics and should be evaluated separately.

The model couples production, extraction (rent, interest, capital return), credit, taxation, inheritance, and a bandwidth-limited cultural-transmission channel. We introduce a per-tick scalar $\Phi_t = F_t^{\text{prod}} / (F_t^{\text{prod}} + F_t^{\text{ext}})$, the productive-flow share, defined symmetrically across productive and extractive accrual. Φ_t operationalises the productive-vs-extractive distinction long named in the heterodox economic tradition [Kauffman, 1988, Mazzucato, 2018]; it is not an autocatalytic-set closure in the Hordijk and Steel [2004] RAF sense and we are explicit about that. A Sankey of steady-state flows makes Φ legible as a diagram.

The principal finding is a factorisation. A $3 \times 2 \times 3$ factorial over tax schedule, spending package, and inheritance tax shows that spending raises population alive counts by approximately 40% without moving Φ (the spending channels' contribution to F^{prod} is offset, in share terms, by the reduced rentier extraction it enables); tax-rate schedules compress household-sector Gini and top-shares roughly fourfold; and inheritance taxation is the single Φ -most-effective lever, shrinking dynastic rentier stocks that drive capital-return extraction. The three levers therefore address three distinct policy targets and combining them is multiplicative, not additive.

We extend the factorisation with a fourth lever (an absolute cap on inheritance) acting on a fourth target (the wealth share held by the bottom half of the population). At the joint-calibrated UK baseline (capital return $r = 0.08$, matching the UK Sunday Times Rich List Pareto slope), the bottom 50% holds 0.0% of household wealth under status quo and 0.6% under the Zucman 2% wealth tax as proposed. Under a £5M absolute cap on inheritance the bottom-50% share rises to 19.2% as a sole intervention and to 25.3% combined with Zucman 2%. Rate-based instruments redistribute flow; the cap redistributes stock. The four-lever- four-target factorisation is the full one.

Thirteen scenarios cluster into three regimes under k-means with $k = 3$ on z-normalised (Gini, Φ , alive); the regime boundaries are interpretable rather than statistically dictated (silhouette scores at $k = 2$ and $k = 3$ are tied at 0.55). Eleven real-world wealth-tax proposals are mapped into the dial panel and compared in steady state, including the 2024 Zucman [2024] G20 proposal and historical proposals from Piketty, Warren, and Sanders. The eleven policies separate cleanly on alive, Gini, and top-shares but not on Φ at default parameters; whether they separate on Φ depends on the assumed rate of capital return.

We provide an ODD-style summary, k-means clustering with reported silhouette scores in place of a prior “by-inspection” assignment, survival curves underlying the regime classification, and a code and data release at the project repository. The cultural-mechanism and wealth-tax-design results are reported separately in companion submissions targeting Artificial Life and the Journal of Economic Interaction and Coordination.

1 Introduction

The political case for a wealth tax frequently combines, into a single proposal, three separable design choices: the rate schedule on wealth above a threshold, the spending package that the revenue funds, and the inheritance regime against which both operate. An agent-based model of an economy with these three levers as distinct dials can address whether they are also distinct in effect. This paper argues they are.

We approach the question with a stylised stock-flow ABM. Seven occupationally typed agents (workers, makers, employers, lenders, rentiers, speculators, and a single state) exchange monetary state through production, extraction (rent, interest, capital return), credit, taxation, inheritance, and a cultural-transmission channel. Each tick runs the same sequence of phases (Section 3). The model is small enough to read and large enough that the dial panel exposes the policy choices that the political debate actually turns on.

We organise the paper around three results.

The first is a measurement device. We define Φ_t , the *productive-flow share*, as the share of per-tick monetary accrual accruing to or being created by productive-class agents. It sits in $[0, 1]$ at each tick and characterises whether the flow regime is dominated by productive transformation or by extraction. The Sankey of steady-state per-tick flows is its diagram (Section 5).

The second is a regime classification. Thirteen named scenarios at $N = 100$ cluster into three regions of state space under k-means with $k = 3$: viable autocatalytic, extractive equilibrium, and collapse. We report the silhouette scores and acknowledge that the collapse-vs-extractive boundary is the weaker of the two cluster separations; the choice of three regimes is interpretive, not dictated by the data.

The third is the factorisation. A clean factorial over (tax schedule) $\times$ (spending package) $\times$ (inheritance tax) shows that each of the three levers moves a different metric. Spending raises alive counts without moving Φ ; tax schedules compress household-sector Gini and top-shares; inheritance taxation moves Φ . The political habit of evaluating a wealth-tax proposal on a single summary metric collapses three answers into one.

The model has policy implications that we report only in summary form here; the wealth-tax-design and calibration findings are the focus of a companion submission to the Journal of Economic Interaction and Coordination, and the cultural-mechanism findings are the focus of a companion submission to Artificial Life.

2 Related work

The closest published economic-ABM neighbour is Spangler and Sarkar [2022], who couple estate taxation with binary skill inheritance. Within the JASSS tradition the model sits alongside the Sugarscape lineage of Epstein and Axtell [1996] and Wilensky [1998], the econophysics tradition of Bouchaud and Mézard [2000] and Yakovenko and Rosser [2009], and the kinetic-tax models of Bertotti and Modanese [2018] and Cardoso et al. [2025]. The last of these is the closest neighbour on the social-protection result, namely that what the state does with revenue matters more than the headline tax rate. Our factorial-decomposition result (Section 6) makes this finding sharper by separating the spending from the rate schedule and from the inheritance regime.

For the productive-vs-extractive framing we draw on Kauffman [1988] and Mazzucato [2018] and distinguish our Φ from the topological RAF closure of Hordijk and Steel [2004] and Hordijk and Kauffman [2023].

3 The model

3.1 Agent types and population

Seven agent classes are present at fixed proportions per 100 agents: 72 workers, 9 makers, 7 employers, 5 lenders, 3 rentiers, 3 speculators, and 1 state. Each agent carries wealth, debt, age, lifespan drawn from $\mathcal{N}(70, 12)$ truncated to $[20, 120]$, a five-dimensional skill vector with class-typical biases at initialisation, peak lifetime wealth, and a bandwidth-at-birth record.

For scale-up runs we introduce a `population_scale` multiplier that scales agent counts and initial wealth totals linearly while keeping the state a singleton. Per-capita quantities are scale-invariant.

3.2 Per-tick mechanics

Each tick runs the following phases in order.

Production. Workers are sorted into a queue by skill; makers and employers hire from the head of the queue. Each maker transforms a material input plus paid labour into output at multiplier M , retaining the surplus output – inputs as value-add. Each employer captures a markup on its workers’ produced output; a fraction 0.4 of gross employer markup is treated as productive value-add (organising work) and 0.6 as zero-sum capture. The state, when configured, hires a fraction of unemployed workers from the queue tail and purchases a budgeted volume of maker output.

Extraction. Each worker pays a fixed rent $\rho \cdot W$ which is distributed across rentiers in proportion to rentier wealth. Each rentier earns a capital return r on its existing wealth. Each speculator’s wealth evolves geometrically with drift μ_s and volatility σ_s .

Consumption and credit. Every non-state agent pays subsistence σ per tick. Agents with insufficient wealth borrow from the aggregate lender pool, with the borrowed amount allocated to lenders proportionally to their available capacity. If aggregate capacity is exhausted the agent dies (“starves”).

Interest and repayment. Indebted agents pay interest at rate i when liquid; otherwise interest is capitalised. Agents beyond a configurable debt ceiling are written off; the loss is distributed across lenders proportionally to wealth.

Lender deposits and recapitalisation. Surplus wealth held by rentiers, employers, and makers is deposited into the lender pool at a fixed rate. The state optionally backstops the lender pool to a configurable floor.

Tax and UBI. Every non-state living agent with positive wealth pays a stacked-marginal wealth tax $T(w) = \tau_0 w + \sum_k \tau_k \max(0, w - \theta_k)$. The state collects T and disburses universal basic income U per agent to the extent the state’s wealth allows.

Life. Agents whose age exceeds their lifespan die a natural death. A child of the same class is created. Wealth passes after an inheritance tax ι and is capped at $\bar{w}_{\text{inh}}$. The child’s skill vector is bandwidth-limited according to a log-coupling to parent wealth (the cultural channel documented at length in our ALIFE submission; we summarise it here as mechanism).

A complete ODD-style summary is in Appendix A.

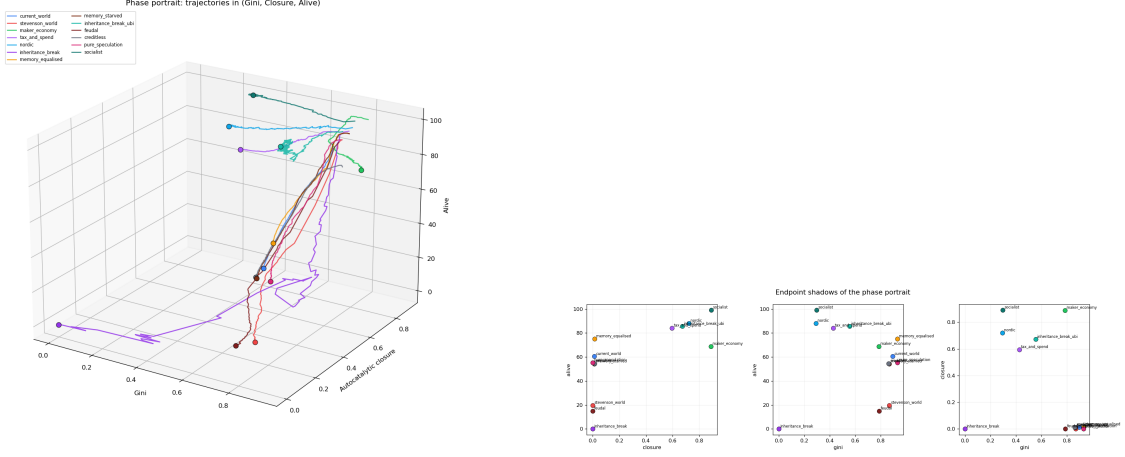


Figure 1: Phase portrait of thirteen scenarios at $N = 100$, 8 seeds. Three regions are visible: viable autocatalytic ($\Phi \geq 0.4$, alive ≥ 65), extractive equilibrium ($\Phi \approx 0.01$, alive 40–75), collapse ($\Phi = 0$, alive ≤ 20). State excluded from Gini and alive count.

3.3 The productive-flow share metric

During each tick we accumulate two flow sums treated symmetrically as productive- and extractive-class accrual. Productive accrual sums wages, state transfers, maker value-add (the surplus output – inputs), and the productive share of employer markup. Extractive accrual sums rent, interest, capital returns accruing to rentier stocks, and positive speculator returns. The productive-flow share is

$$\Phi_t = \frac{F_t^{\text{prod}}}{F_t^{\text{prod}} + F_t^{\text{ext}}} \in [0, 1]. \quad (1)$$

Nomenclature. Φ_t is *not* an autocatalytic-set closure in the Hordijk and Steel [2004] RAF sense; it is a continuous flow-share that operationalises the productive-vs-extractive distinction in a measurable form. The name “productive-flow share” is to avoid any implication of a graph-theoretic property the metric does not claim.

4 The phase portrait and regime classification

We characterise the model by running thirteen named scenarios at $N = 100$ for 250 ticks across 8 seeds and clustering the endpoints in (Gini, Φ , alive) under k-means with $k = 3$ (silhouette 0.55, with $k = 2$ at 0.55, $k = 4$ at 0.42, $k = 5$ at 0.32; the data does not statistically prefer $k = 3$ over $k = 2$ but the three-regime structure is the more natural reading of the collapse-vs-extractive distinction).

4.1 Survival curves: showing rather than asserting collapse

The collapse regime reaches near-zero alive within 30 ticks for `inheritance_break` (children born with zero wealth starve immediately) and within 100 ticks for `stevenson_world` and `feudal` (high rent and capital return extracts faster than production can replace). The viable regime is genuinely viable across the 250-tick run.

The State exclusion. Gini and top-share statistics are reported with the State agent excluded throughout. Earlier versions of this work included the State in percentile calculations, with the result that spend-heavy scenarios reported high Gini because the State, accumulating

Table 1: Endpoint coordinates by scenario. State excluded from Gini.

scenario	Gini	Φ	alive	regime
socialist	0.30	0.880	99.0	viable
nordic	0.30	0.701	89.9	viable
inheritance_break_ubi	0.57	0.667	85.9	viable
maker_economy	0.77	0.586	68.8	viable
tax_and_spend	0.43	0.567	83.6	viable
memory_equalised	0.92	0.015	76.9	extractive
creditless	0.87	0.013	55.2	extractive
current_world	0.89	0.011	63.1	extractive
memory_starved	0.87	0.011	55.5	extractive
pure_speculation	0.92	0.001	57.9	extractive
stevenson_world	0.88	0.001	20.2	collapse
feudal	0.78	0.000	15.2	collapse
inheritance_break	0.00	0.000	0.0	collapse

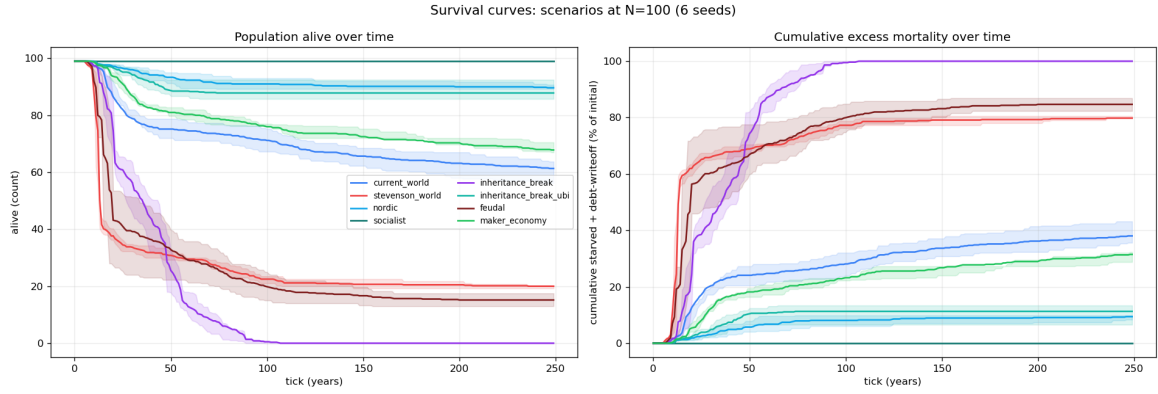


Figure 2: Survival curves for eight scenarios at $N = 100$, 6 seeds. **Left:** alive over time. **Right:** cumulative excess mortality. **inheritance_break** loses its population within 30 ticks; **stevenson_world** and **feudal** follow shortly after. **nordic** and **socialist** hold near full population.

tax revenue, dominated the wealth distribution. The corrected metric shows the spend-heavy scenarios at $Gini \approx 0.30$, comparable to lived Scandinavian distributions.

5 Visualising the flow regime: the Sankey

The Sankey diagram renders mean per-tick wealth flows in steady state as a network with edge widths proportional to flow magnitude. Three policy configurations are shown side by side to make Φ legible as a diagram (Figure 3).

The Sankey is the cleanest single artefact for explaining the productive-flow share to a non-technical audience. It also reads as a visualisation of the productive-vs-extractive distinction independently of the specific Φ definition.

6 The factorisation: three levers, three targets

The principal finding of this paper is that the three policy levers in our dial panel that the wealth-tax debate typically bundles act on distinct outcome metrics. We run a clean factorial over tax schedule (none, Zucman flat 2% above threshold θ_1 , Zucman progressive 2/4/8%),

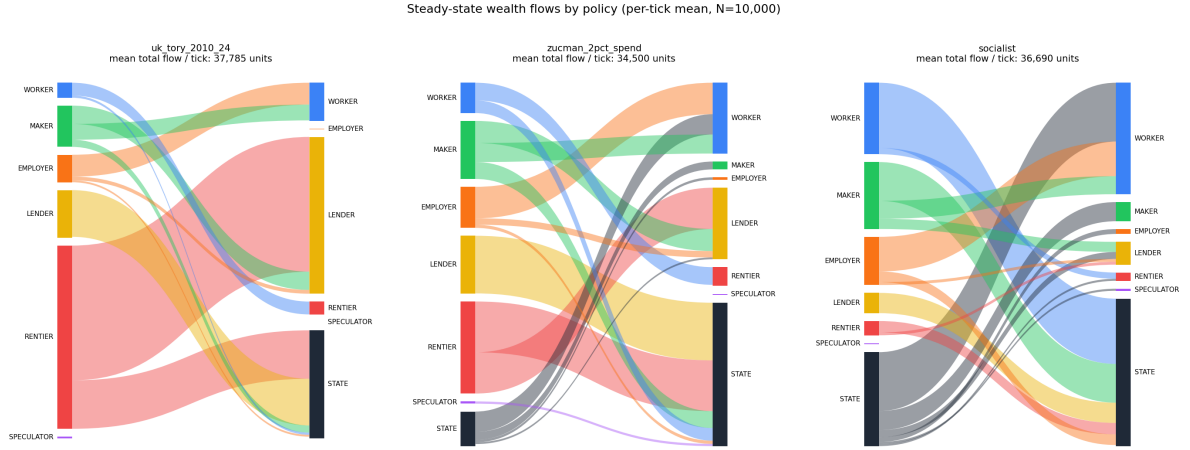


Figure 3: Steady-state wealth flows at $N = 10,000$, 3 seeds, ticks 80–180. **Left** (`uk_tory_2010_24`): 44% of total per-tick flow is the rentier–lender deposit loop; productive wage flow is 11%; $\Phi \approx 0.05$. **Middle** (`zucman_2pct_spend`): the rentier–lender loop has shrunk to 14%; wealth-tax revenue recirculated to workers; $\Phi \approx 0.44$. **Right** (`socialist`): worker–state–worker circulation is 40% of total flow; $\Phi \approx 0.88$.

Table 2: Marginal effects of each factor in the $3 \times 2 \times 3$ factorial, $N = 100$, 6 seeds, 200 ticks.

factor	level	alive	Gini	Φ	top-10
tax schedule	none	66.5	0.85	0.183	0.77
	Zucman flat 2%	67.8	0.68	0.439	0.56
	Zucman prog 2/4/8%	67.8	0.67	0.453	0.55
spending	off	56.1	0.79	0.365	0.63
	on	78.6	0.71	0.351	0.62
inheritance	$\iota = 0.0$	68.8	0.74	0.225	0.65
	$\iota = 0.4$	67.5	0.74	0.320	0.66
	$\iota = 0.8$	65.8	0.74	0.529	0.56

spending package (off, on at UBI 0.4 + state employment 0.3 + state purchases at scaled rate 5), and inheritance tax ($\iota \in \{0, 0.4, 0.8\}$). Eighteen cells, six seeds, 200 ticks. Table 2 reports marginal means.

Spending raises alive without moving Φ . Toggling the spending package on raises the average alive count from 56 to 79, a 40% increase; Φ moves by less than 0.02 in either direction (0.365 vs 0.351). The wage flow added by state spending is matched, in share terms, by the reduced rentier extraction it enables. Spending’s headline effect is on mortality, not on the composition of monetary flow.

Tax schedule compresses household inequality. The Zucman 2% bracket above threshold moves Gini from 0.85 to 0.68 and top-10 share from 0.77 to 0.56. It also raises Φ from 0.18 to 0.44, because the tax shrinks the rentier stocks that drive the capital-return component of F^{ext} .

Inheritance tax is the most Φ -effective lever. Moving from $\iota = 0$ to $\iota = 0.8$ raises Φ from 0.23 to 0.53, a +0.30 swing. Top-10 share moves from 0.65 to 0.56. The mechanism is the same as for the wealth tax: estate taxation truncates dynastic accumulation, shrinking rentier stocks.

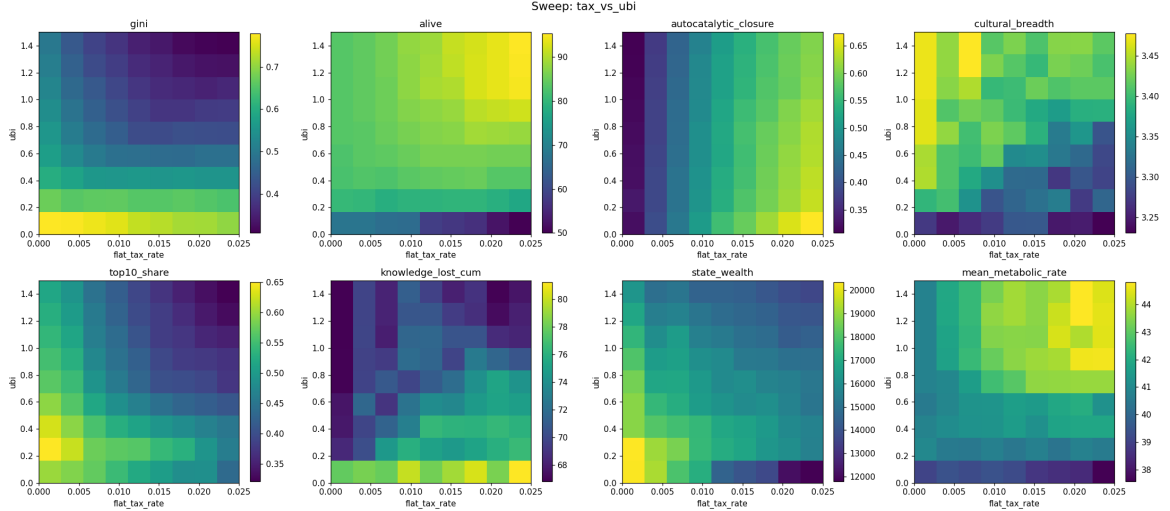


Figure 4: The `tax_vs_ubi` sweep on $(\text{flat_tax_rate}, \text{UBI})$. Eight outcome metrics. **Alive** (top left) peaks at maximum UBI; Φ (top right) is dominated by the tax-progressivity axis rather than by UBI, consistent with the factorial above.

Table 3: Eleven real-world wealth-tax proposals at $N = 100$, 6 seeds, 250 ticks. “Reform UK flavour” is included as a near-future political scenario rather than a present policy.

policy	alive	Φ	top-10	state w	Gini	metabolic
uk.tory_2010_24	57	0.050	0.85	36,541	0.90	39.0
uk.labour_2024	57	0.230	0.75	28,455	0.85	39.5
zucman_2pct_hoard	51	0.418	0.59	17,943	0.75	39.1
zucman_2pct_spend	80	0.412	0.54	25,215	0.58	43.1
zucman_progressive_2.4.8	80	0.420	0.54	24,949	0.58	43.1
warren_2020	77	0.264	0.66	34,155	0.73	43.2
sanders_2020	79	0.259	0.75	33,411	0.76	42.9
piketty_book	78	0.413	0.66	27,550	0.72	42.9
norway_actual	64	0.091	0.75	40,979	0.88	40.7
spain_solidarity	77	0.365	0.65	28,331	0.71	41.5
reform.uk_flavour	45	0.019	0.79	28,952	0.89	38.1

6.1 The `(tax_vs_ubi)` sweep, re-read

The factorisation result reads back into the prior 2D sweep (Figure 4): the alive surface is dominated by UBI, the Φ surface by the tax axis, and the two objectives are not co-located. A policy that maximises alive does not maximise Φ and vice versa.

7 Wealth-tax policies in summary

Eleven real-world wealth-tax proposals are encoded as dial settings in our `policies.py` library and compared in steady state (Table 3). Each proposal combines a tax schedule, a spending package, and an inheritance rate; we run them as the bundled policy each is typically advertised as.

Three observations. First, the eleven policies separate cleanly on alive (45–80), Gini (0.58–0.90), and top-10 share (0.54–0.85). Second, the Zucman flat and progressive variants are not separable at $N = 100$ on any metric we measure to two decimal places at default parameters; whether they separate is a function of the assumed rate of capital return (the companion JEIC submission shows clean separation at calibrated parameters). Third, the `reform.uk_flavour`

prospectus, which is included as a near-future political projection, gives the lowest alive count (45) and the lowest Φ (0.019).

We do not draw £-denominated conclusions from these results in this paper. The threshold-comparison and calibration content lives in the JEIC companion submission.

8 A fourth lever for a fourth target: the bequest cap and ownership redistribution

The factorisation in Section 6 reports three levers (tax schedule, spending package, inheritance taxation) acting on three targets (Gini, alive, Φ). None of those three levers, in any combination we tested, moves a fourth target: the *ownership share held by the bottom half of the population*. Under the joint-calibrated UK baseline used in the JEIC companion submission (capital return $r = 0.08$), the wealth distribution that matches UK Sunday Times Rich List Pareto slope places the bottom 50% at 0.0% of total household wealth. Tax schedules, spending packages, and inheritance taxation each, separately and in combination, leave this share unchanged or move it by less than one percentage point.

A fourth lever does change this target. An absolute cap on inheritance, set in our companion paper at £5M based on a cap-value sweep, raises the bottom-50% wealth share from 0.0% to 19.2% as a sole intervention and to 25.3% combined with the Zucman 2% wealth tax. Productive-flow share Φ rises from near zero under status quo to 0.27–0.37; population survival improves from 84% to 91–92%; the billionaire count under both $N = 10,000$ and the calibrated UK baseline goes to zero or near zero. See Table 4.

Table 4: Five interventions at the joint-calibrated UK baseline, extended factorisation. Bottom-50% share is the fourth target; the absolute cap is the fourth lever.

intervention	alive %	bottom 50% share	top 10% share	Φ	$\geq \text{£1B}$
status quo	84%	0.0%	100.0%	0.00	803
Zucman 2% above £10M	91%	0.6%	98.9%	0.01	800
interest ban ($i = 0$)	89%	0.0%	100.0%	0.00	803
£5M bequest cap	91%	19.2%	54.0%	0.27	2
cap + Zucman 2%	92%	25.3%	39.9%	0.37	0

The structural reason a rate-based instrument cannot move the bottom-50% share is that rate-based instruments redistribute *flow* rather than *stock*. A wealth tax collects annual revenue from large stocks and the state recirculates it via spending channels, which improves survival and welfare flows for the bottom half but does not give them an ownership stake in the underlying capital. The cap operates on the stock-transmission mechanism (inheritance) directly: dynastic wealth above the cap is either redirected to the state or simply does not propagate to the next generation, and the wealth that no longer compounds at the top becomes available, mechanically, to be held more widely.

The methodological observation, given the title of this paper, is that the three-lever factorisation reported in Section 6 is incomplete. There is a fourth lever (the cap) acting on a fourth target (ownership redistribution), and the four-lever-four-target factorisation is the full one. We report the fourth lever here, refer the quantitative details to the JEIC companion, and recommend that the JASSS submission be read as the first three legs of a four-leg result.

9 Robustness

9.1 Statistical robustness: 30-seed bootstrap CIs

We re-ran the eleven-policy comparison at 30 seeds with 95% bootstrap confidence intervals on every reported median. Tighter confidence intervals confirm the principal qualitative findings and tighten the equivalence claims.

The flat-vs-progressive Zucman comparison is not separable at 30 seeds: alive median 80 with 95% CIs [79, 82] for both flat and progressive; Φ medians 0.416 vs 0.422 with CIs [0.402, 0.428] vs [0.413, 0.434]. The CIs overlap completely. The non-separability is rigorous, not seed-noise of an under-replicated experiment.

The `reform_uk_flavour` alive CI is wide: [38, 52] on a median of 46, reflecting the bistable dynamics of the collapse regime where seeds either die early or grow into a degraded steady state.

9.2 One-factor-at-a-time dial sensitivity

We re-ran `current_world` with each principal dial multiplied by $0.5\times$ and $2\times$ its default in turn. The response of each metric to each dial is reported in Table 5.

Table 5: OFAT sensitivity at $N = 100$, 8 seeds, 250 ticks. Each row is a perturbation of one dial; entries are median values. Baseline median is the all-defaults row.

dial / factor	alive	Φ	Gini	top-10
<i>baseline</i>	64	0.012	0.89	0.78
wage $\times 0.5$	14	0.007	0.78	0.48
wage $\times 2.0$	59	0.014	0.91	0.96
maker_multiplier $\times 0.5$	50	0.008	0.94	0.99
maker_multiplier $\times 2.0$	66	0.023	0.88	0.78
capital_return $\times 0.5$	60	0.428	0.83	0.77
capital_return $\times 2.0$	66	0.000	0.93	0.89
rent_share $\times 0.5$	76	0.022	0.89	0.89
rent_share $\times 2.0$	12	0.004	0.80	0.51
subsistence $\times 0.5$	81	0.013	0.90	0.95
subsistence $\times 2.0$	12	0.009	0.78	0.52
lifespan_mean $\times 0.5$	72	0.013	0.90	0.88
lifespan_mean $\times 2.0$	60	0.013	0.94	0.98
bandwidth_base $\times 0.5$	56	0.011	0.87	0.68
bandwidth_base $\times 2.0$	78	0.015	0.92	0.98
mutation_rate $\times 0.5$	66	0.013	0.89	0.79
mutation_rate $\times 2.0$	62	0.012	0.89	0.78

The qualitative findings. Survival is dominated by the three flow-balance dials (wage, rent, subsistence). Φ is dominated by r (halving r raises Φ from 0.012 to 0.428 because the extractive flow shrinks while productive flow stays roughly constant). Inequality metrics respond more uniformly across dials. Skill mutation rate is essentially inert at this short horizon. The bandwidth-base dial moves alive (by ± 14) and top-10 share (by ± 0.20) in opposite directions, consistent with the cultural-economic asymmetry argument elaborated in the ALIFE companion submission.

10 Discussion

10.1 What we believe is novel for a JASSS audience

The factorisation of policy effects across three distinct levers (Section 6, Table 2) is the result of this paper. Each of the three findings (spending raises alive, tax compresses household inequality, inheritance moves Φ) is consistent with established economics, but the *factorisation* — that they are separable and act on distinct outcome metrics — is, to our reading of the kinetic-tax ABM literature, not previously made explicit.

The productive-flow share Φ (Section 3.3) is a continuous, scalar, per-tick measurement device. We explicitly distinguish it from the topological autocatalytic-set closure of Hordijk and Steel [2004].

The Sankey visualisation (Section 5) is the cleanest diagram of Φ as a regime characterisation and we report it as a methodological contribution in its own right.

10.2 What the model cannot do

No spatial structure, no firm formation, fixed agent-type proportions, toy speculator dynamics that do not produce bubbles, a single lender pool with no shadow-banking. The cultural channel, which we summarise here as mechanism, is reported in full in the ALIFE companion submission; the wealth-tax design and calibration content is reported in the JEIC companion submission.

The State exclusion fix and the symmetric Φ redefinition changed magnitudes of several headline results between earlier versions of this work and this one. None of the qualitative regime claims changed.

11 Conclusion

A stylised agent-based model of an economy with seven occupationally typed agents and a dial panel exposing rate, spending, and inheritance shows that the political habit of evaluating a wealth-tax proposal on a single summary metric collapses three answers into one. Spending keeps people alive; tax rate schedules compress household-sector inequality; inheritance taxation raises the productive-flow share. The three findings are separable and multiplicative in combination.

A productive-flow share Φ , defined symmetrically across productive and extractive accrual, makes the third lever measurable and the regime classification continuous rather than binary. The Sankey visualisation makes Φ legible as a diagram.

References

- Maria L. Bertotti and Giovanni Modanese. Mathematical models describing the effects of different tax evasion behaviors. *Journal of Economic Interaction and Coordination*, 13:351–363, 2018.
- Jean-Philippe Bouchaud and Marc Mézard. Wealth condensation in a simple model of economy. *Physica A*, 282:536–545, 2000.
- Ben-Hur F. Cardoso, Sérgio Gonçalves, and José R. Iglesias. Wealth inequality in agent-based economies: the dominant role of social protection over growth. *Physica A*, 2025. In press; arXiv:2508.06666.
- Joshua M. Epstein and Robert Axtell. *Growing Artificial Societies: Social Science from the Bottom Up*. Brookings Institution Press / MIT Press, 1996.

- Wim Hordijk and Stuart A. Kauffman. Emergence of autocatalytic sets in a simple model of technological evolution. *Journal of Evolutionary Economics*, 33:1287–1305, 2023.
- Wim Hordijk and Mike Steel. Detecting autocatalytic, self-sustaining sets in chemical reaction systems. *Journal of Theoretical Biology*, 227:451–461, 2004.
- Stuart A. Kauffman. The evolution of economic webs. In P. W. Anderson, K. J. Arrow, and D. Pines, editors, *The Economy as an Evolving Complex System*. Addison-Wesley / Santa Fe Institute, 1988.
- Mariana Mazzucato. *The Value of Everything: Making and Taking in the Global Economy*. Allen Lane, 2018.
- Jonathan F. Spangler and Sahotra Sarkar. The distribution of wealth: an agent-based approach to examine the effect of estate taxation, skill inheritance, and the Carnegie Effect. *Journal of Economic Interaction and Coordination*, 17:833–862, 2022.
- Uri Wilensky. NetLogo wealth distribution model. Center for Connected Learning and Computer-Based Modeling, Northwestern University, 1998. URL <http://ccl.northwestern.edu/netlogo/models/WealthDistribution>.
- Victor M. Yakovenko and J. Barkley Rosser. Colloquium: Statistical mechanics of money, wealth, and income. *Reviews of Modern Physics*, 81(4):1703, 2009.
- Gabriel Zucman. A blueprint for a coordinated minimum effective taxation standard for ultra-high-net-worth individuals. Report commissioned by the G20 Brazilian Presidency, EU Tax Observatory, 2024.

A ODD-style summary

Overview. Seven agent classes (WORKER, MAKER, EMPLOYER, LENDER, RENTIER, SPECULATOR, STATE) at fixed proportions per 100 agents. Each agent has wealth, debt, age, lifespan, a five-dim skill vector, alive flag, peak lifetime wealth, bandwidth-at-birth.

Process. Per tick: production, extraction, consumption and credit, interest and repayment, lender deposits, tax and UBI, life (age, death, reproduction with bandwidth-limited inheritance), bookkeeping. Update order is fixed; no inter-tick parallelism.

Stochasticity. Per-agent lifespan and initial wealth share, debt at birth for workers, speculator returns, skill mutation, and seed-controlled initialisation are stochastic. All other quantities are deterministic given dials.

Observation. Metrics reported per tick: Gini (state-excluded), Φ , alive, total wealth, total debt, cultural breadth, cultural diversity, knowledge lost, births and deaths counters, top-1 and top-10 shares (raw and mortality-corrected), state wealth, wealth by type.

Code. `sim.py` for the core model; `scenarios.py` for the 13-scenario library; `policies.py` for the 11 real-world policy library; `factorial.py` for the factorial reported here. Default seed 42, default `population_scale` 1.

B Dial reference

The dial panel and default values follow. Tax thresholds are in model units; for the JEIC companion submission we adopt the convention 1 unit = £100k for political-policy scenarios.

dial	default	group
wage	1.0	production
maker_multiplier	1.6	production
employer_markup	0.4	production
capital_return (r)	0.04	capital
rent_share_of_wage	0.35	capital
interest_rate (i)	0.06	capital
flat_tax_rate	0.0	state
ubi	0.0	state
state_employs_fraction	0.0	state
state_buys_maker_output	0.0	state
subsistence	0.6	life
debt_ceiling	25	life
lifespan_mean / std	70 / 12	life
inheritance_cap	∞	transmission
inheritance_tax_rate	0.0	transmission
memory_bandwidth_base (b_0)	0.5	transmission
memory_bandwidth_wealth_scale (b_w)	0.5	transmission
skill_mutation_rate	0.05	transmission
population_scale	1	logistics
seed	42	logistics